

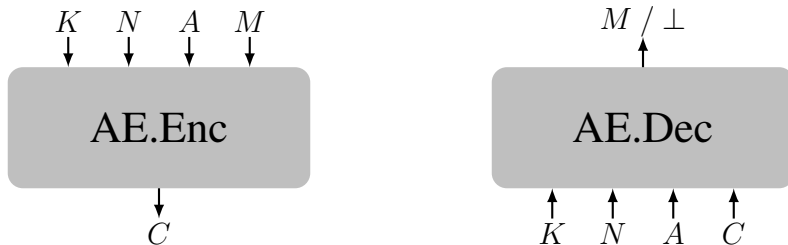
Generic Committing Attacks Zero-Padded ASCON is Less Secure than Expected

Hrithik Nandi

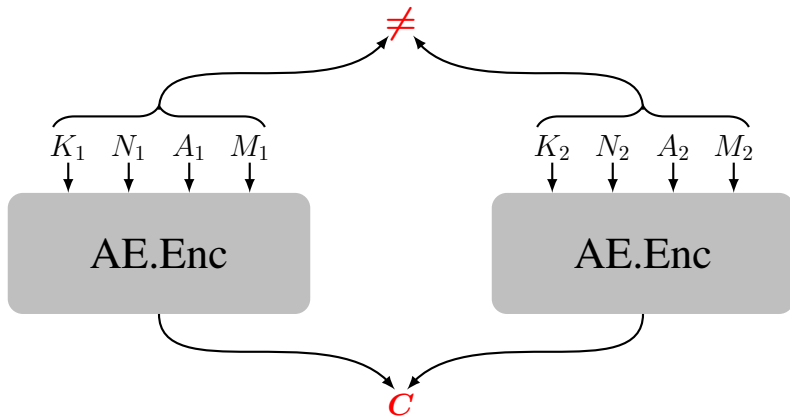
Institute for Advancing Intelligence (IAI), TCG CREST
&
Ramakrishna Mission Vivekananda Educational and Research Institute

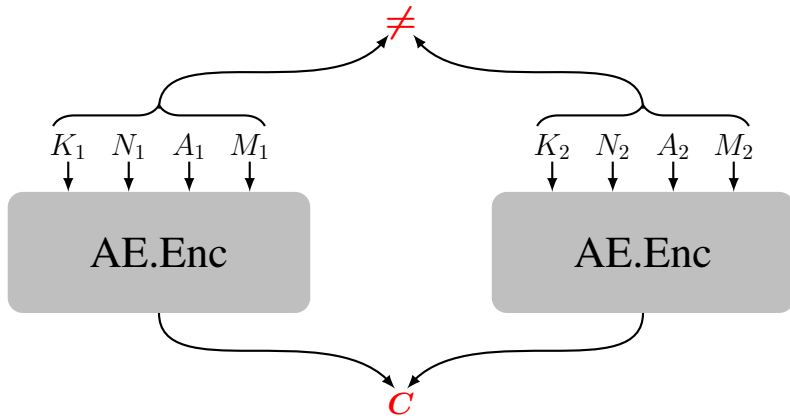


September 24, 2026



- Traditional security requires
Privacy of message M + **Authenticity** of both M and A .
- AEAD provide both integrity and privacy in a single scheme.
- Some protocols need beyond the traditional AE security.





CMT-4

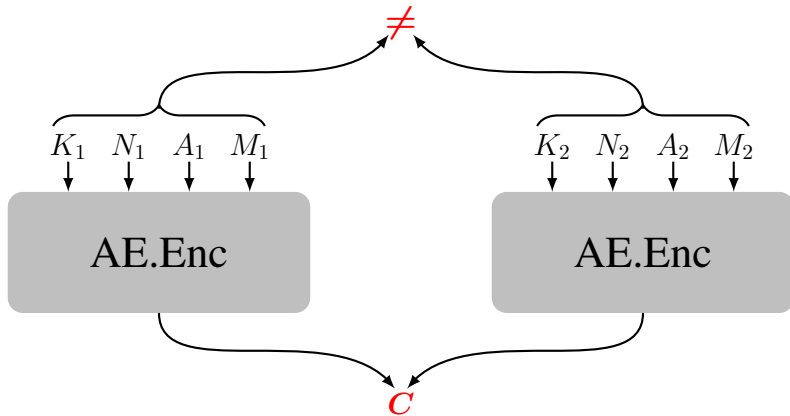
$(K_1, N_1, A_1, M_1) \neq (K_2, N_2, A_2, M_2)$

CMT-1

$K_1 \neq K_2$

CMT- K^*

$K_1 \neq K_2, (N_1, A_1) = (N_2, A_2)$



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Current Scenario of Existing Schemes

- Existing AEADs had not been built to capture the new CMT security Notions.
- Attacks have been given on AES-GCM, OCB3, AES-GCM-SIV, CCM, EAX, SIV etc.
- Most of the NIST LWC finalists have also been attacked.

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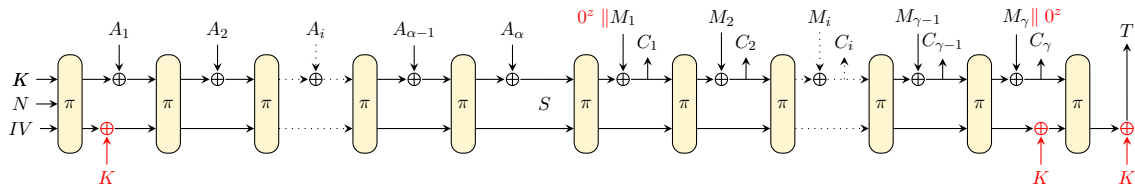
A Simple Fix

- One simple solution was proposed – Zero-padding.
- CMT-K security of GCM increases to $\tau/2$ -bit security by prepending 0^τ in M .
- Two possibilities of appended bit positions: prefix or suffix.

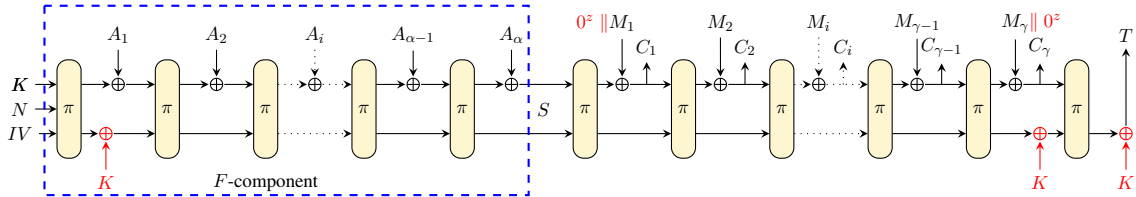


- We treat Duplex and Ascon in an integrated way and consider **Zero-Padding**.
- **Duplex**: A Permutation based construction.

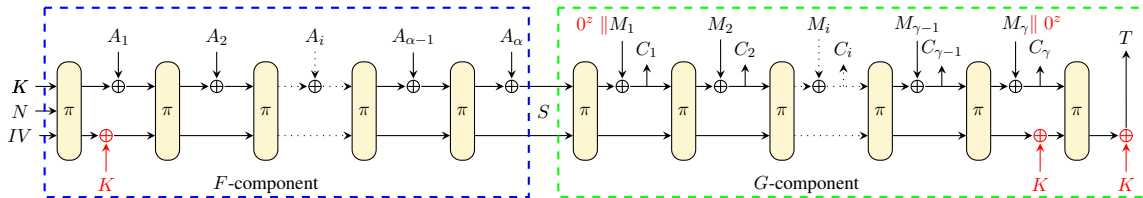
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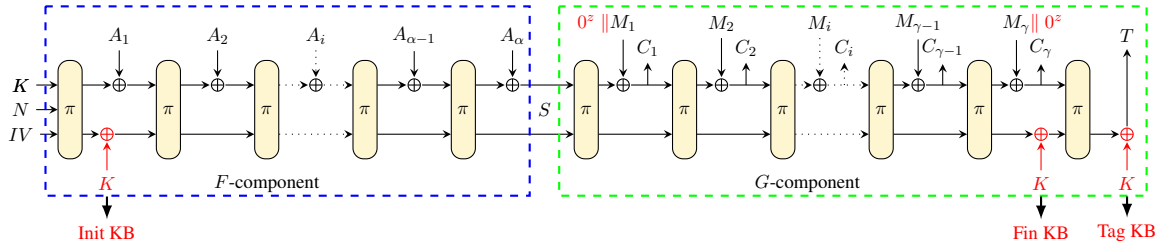
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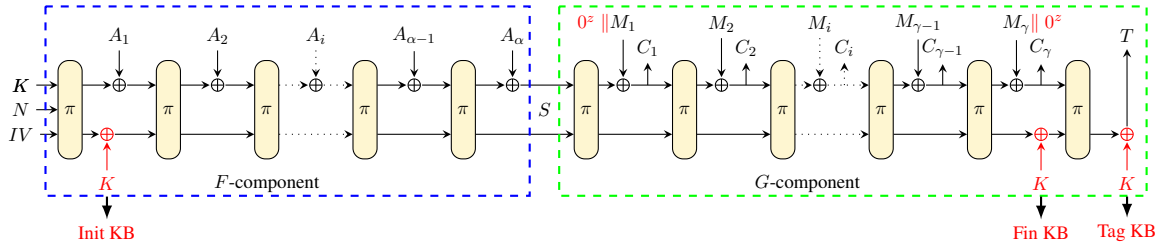
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- **Ascon**: NIST standard for lightweight cryptography (SP800-232).
→ All **three KBs** are used.

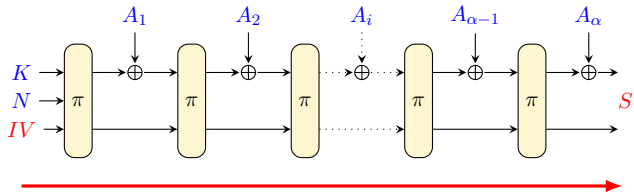


- We give a systematic analysis of existing attack strategies: (i) IS, (ii) CC, and (iii) TC.
- Consider CMT-4 and CMT-K^{*} notions.
- Apply various combinations of KB and ZP.

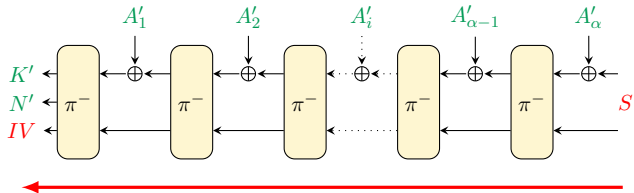
CMT-4 ATTACKS

CMT-4: INITIAL STATE (IS) ATTACK

- Pick an arbitrary context (K, N, IV) and compute S .



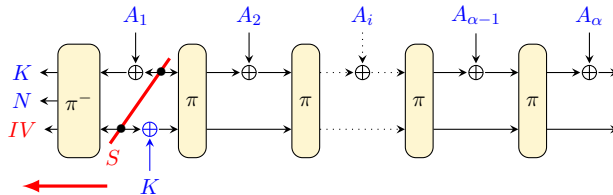
- Invert S using **different** associated data A' until IV is matched.



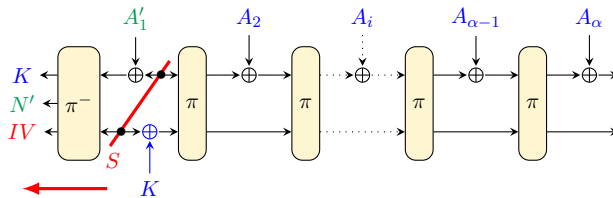
Attack Complexity is 2^l .

CMT-4: INITIAL STATE (IS) ATTACK WITH KB

- Pick an intermediate state S and **invert** it using **different** associated data.



- For this strategy, we require a **collision** for the keys (due to KB).



Attack Complexity is $2^{l+\frac{\kappa}{2}}$.



- The **previous IS attack** on Duplex remains **applicable** even when the scheme **uses ZP** (prefix/suffix).

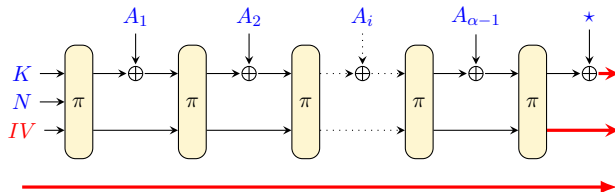


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- The attack **targets the first component F** and ensures that the states outputted by F agree.

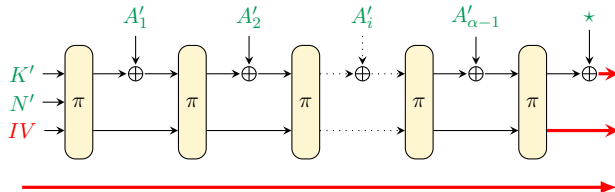


- The **previous IS attack** on Duplex remains **applicable** even when the scheme **uses ZP** (prefix/suffix).
- The attack **targets the first component F** and ensures that the states outputted by F agree.
- The **same zero-padded message** is then used as the input to the second component G , yielding **identical ciphertexts and tags**.

- Use **different** (K, N, A) to obtain an intermediate state S with capacity collision.



- Capacity collision is then **turned into a full-state collision** via the absorption of suitable block.

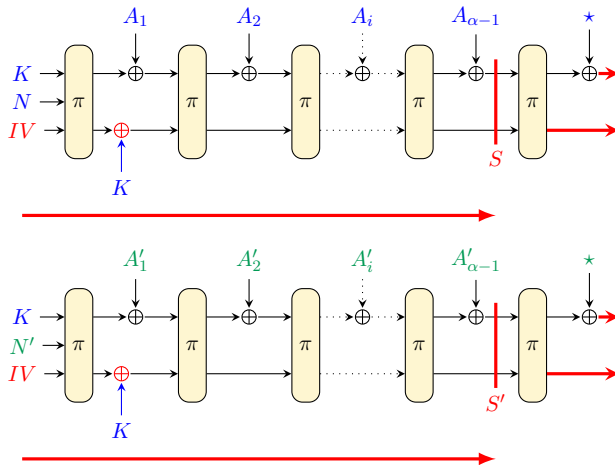


Attack Complexity is $2^{\min(\frac{c_F}{2}, \frac{c_G}{2})}$.

CMT-4: CAPACITY COLLISION (CC) ATTACK WITH KB



- Use same K , **different** (N , A) to obtain an intermediate state S with capacity collision.



Attack Complexity is $2^{\min(\frac{c_F}{2}, \frac{c_G}{2})}$.

- The **CC attack** on Duplex still **works** when the scheme **deploys ZP**.



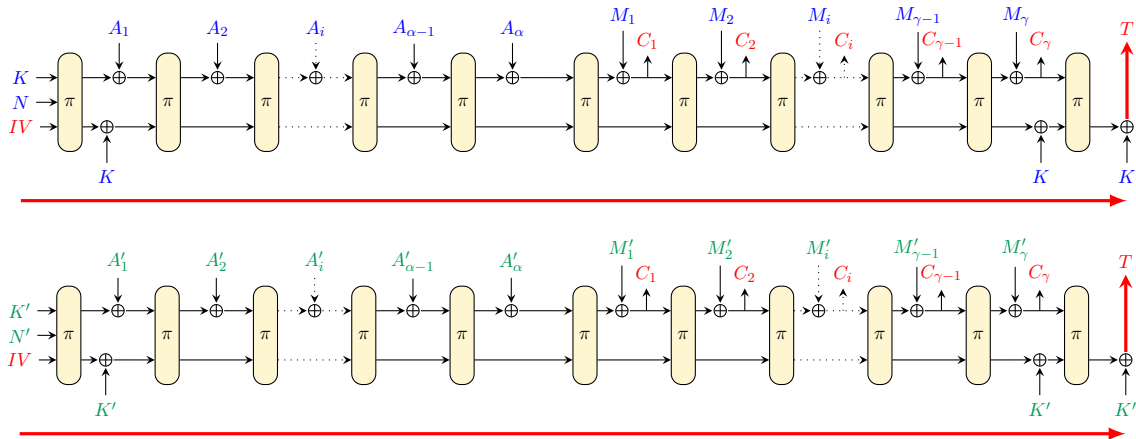
- The **CC attack** on Duplex still **works** when the scheme **deploys ZP**.
- Targeting capacity of an **intermediate state of F**
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- The **CC attack** on Duplex still **works** when the scheme **deploys ZP**.
- Targeting capacity of an **intermediate state of F**
 - Complexity is $\frac{cF}{2}$, irrespective of prefix-ZP or suffix-ZP.
- Targeting capacity of an **intermediate state of G** , complexity differs
 - For **prefix-ZP** complexity is $\frac{cG+z}{2}$, to ensure the collision of ciphertexts on ZP.
 - For **suffix-ZP** complexity is $\frac{cG}{2}$, targetting state should before the zero-padding.

CMT-4: TAG COLLISION (TC) ATTACK (WITH KB)

- Use arbitrary contexts until a collision of the tags is found. The attack is **unaffected** by any KB.



Attack Complexity is $2^{\frac{\tau}{2}}$.



- The TC attack **still applies** in case of **zero-padding** – regardless of prefix or suffix.



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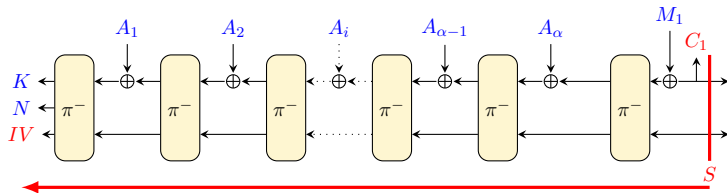
- The TC attack **still applies** in case of **zero-padding** – regardless of prefix or suffix.
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- The attack complexity increases to $\frac{\tau+z}{2}$.

	Cost		
	IS	CC	TC
Duplex[0, 0, 0]	l	$\min\left(\frac{c_F}{2}, \frac{c_G}{2}\right)$	$\frac{\tau}{2}$
Duplex[*, *, *]	$\frac{\kappa}{2} + l$	$\min\left(\frac{c_F}{2}, \frac{c_G}{2}\right)$	$\frac{\tau}{2}$

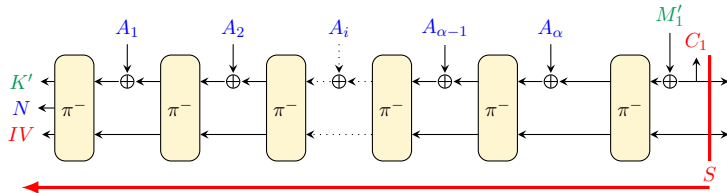
	Cost			
	IS	CC		TC
		Prefix	Suffix	
ZP-Duplex[0, 0, 0]	l	$\min\left(\frac{c_F}{2}, \frac{c_G+z}{2}\right)$	$\min\left(\frac{c_F}{2}, \frac{c_G}{2}\right)$	$\frac{\tau+z}{2}$
ZP-Duplex[*, *, *]	$\frac{\kappa}{2} + l$	$\min\left(\frac{c_F}{2}, \frac{c_G+z}{2}\right)$	$\min\left(\frac{c_F}{2}, \frac{c_G}{2}\right)$	$\frac{\tau+z}{2}$

CMT-K[★] ATTACKS

- We target the intermediate state **after absorbing first message block**.

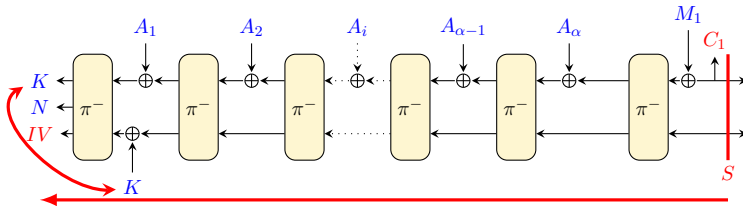


- Invert** S using **different** M'_1 (same A) until N collides and IV is matched.

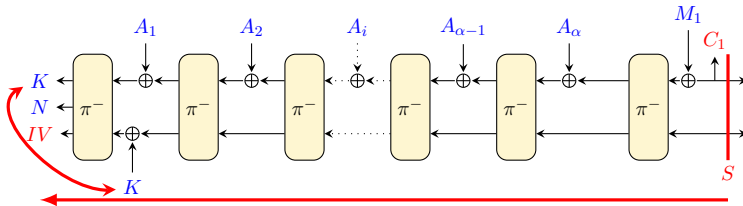


Attack Complexity is $2^{\frac{\nu}{2}+l}$.

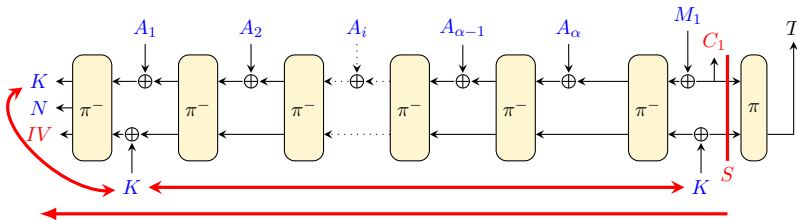
- Init-KB used then the **output key** should **matched** with **the one used in KB** – $2^{\kappa + \frac{\nu}{2} + l}$.



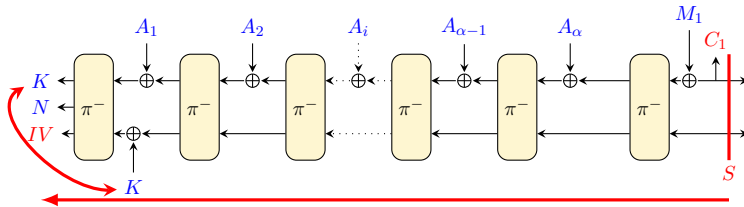
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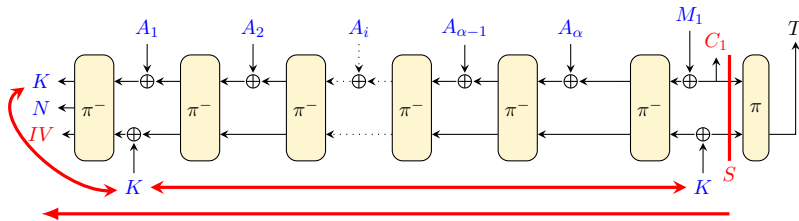
- Same strategy with **only 1 block message** when Fin-KB used – $2^{\kappa+\frac{\nu}{2}+l}$.



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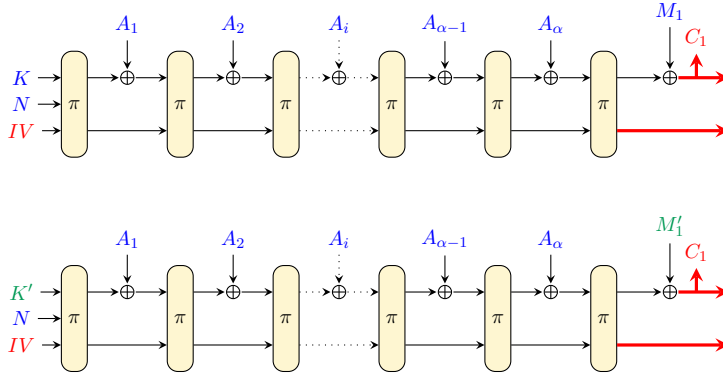


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- Similar attack strategy as described for Duplex works.
- The degree of freedom of the adversary is reduced due to ZP.
- For prefix-ZP the adversary no longer have full control over the first message block.
- For suffix-ZP there are no changes as target state is after the first round message absorption.
But, if Fin-KB used then the DoF is reduced.

- Target is **capacity of the state in the message absorption component**, by using **different keys**.

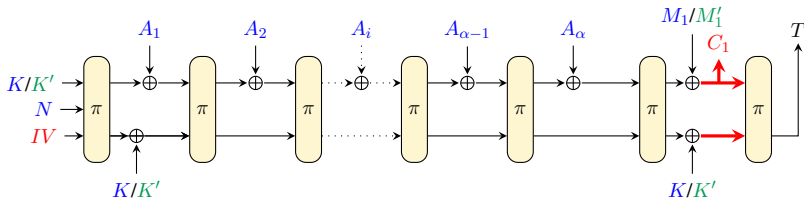


Attack Complexity is $2^{\frac{c_G}{2}}$.



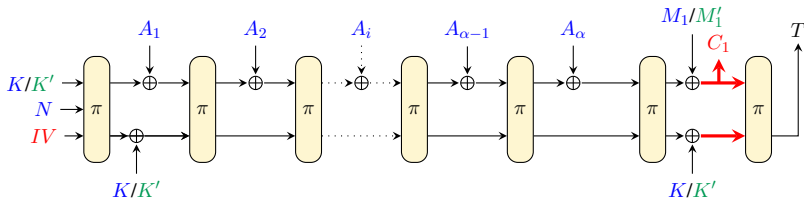
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Attack Complexity is $2^{\frac{c_G}{2}}$.

- If **Tag-KB** used then the attack **does not work** since the attack utilizes different keys.

- The attacks proceed analogously to the case without zero-padding.



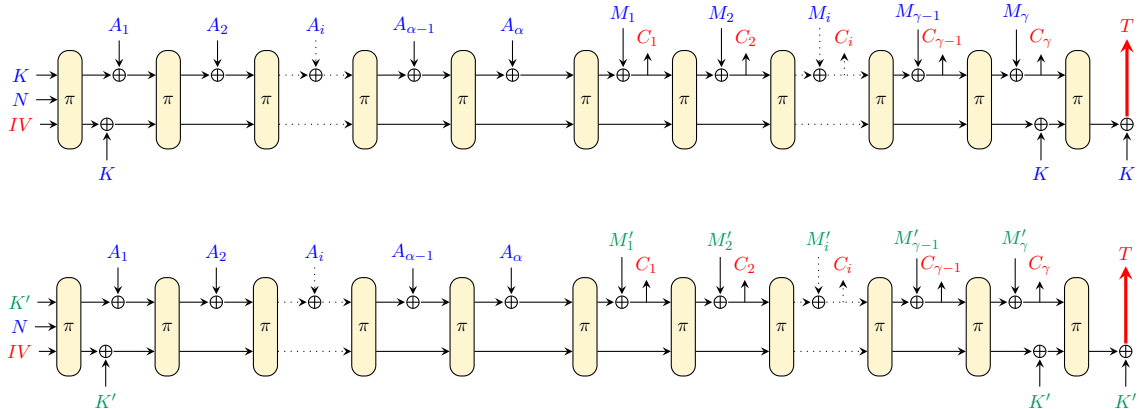
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- For **suffix-ZP** the attacks and their complexities are **exactly same as without ZP**.
- For **suffix-ZP with Fin-KB** the **last z -bits of final ciphertext have to be collide**, the complexity increases to $2^{\frac{c_G+z}{2}}$.

- Similar as CMT-4, but use **different K** same (N, A) . The attack is **unaffected** by any KB.



Attack Complexity is $2^{\frac{\tau}{2}}$.

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- For **suffix-ZP** as the **last** ciphertext must **agree on z -bits**, the complexity increases to $2^{\frac{\tau+z}{2}}$.

	IS		CC		TC	
	Cost	DoF	Cost	DoF	Cost	DoF
Duplex[0, 0, 0]	$\frac{\nu}{2} + l$	r_G	$\frac{c_G}{2}$	κ	$\frac{\tau}{2}$	κ
Duplex[1, 0, 0]	$\frac{\nu}{2} + l + \kappa$	r_G	$\frac{c_G}{2}$	κ	$\frac{\tau}{2}$	κ
Duplex[*, 1, 0]	$\frac{\nu}{2} + l + \kappa$	$r_G + \kappa$	$\frac{c_G}{2}$	κ	$\frac{\tau}{2}$	κ
Duplex[*, *, 1]	—	—	—	—	$\frac{\tau}{2}$	κ

Table: CMT_K^{*} results for duplex sponge construction with prefix zero-padding.

	IS		CC		TC	
	Cost	DoF	Cost	DoF	Cost	DoF
ZP-Duplex[0, 0, 0]	$\frac{\nu}{2} + l$	$r_G - z$	$\frac{c_G + z}{2}$	κ	$\frac{\tau + z}{2}$	κ
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ZP-Duplex[*, *, 1]	—	—	—	—	$\frac{\tau + z}{2}$	κ

Table: CMT_K^{*} results for duplex sponge construction with suffix zero-padding.

	IS		CC		TC	
	Cost	DoF	Cost	DoF	Cost	DoF
ZP-Duplex[0, 0, 0]	$\frac{\nu}{2} + l$	r_G	$\frac{c_G}{2}$	κ	$\frac{\tau + z}{2}$	κ
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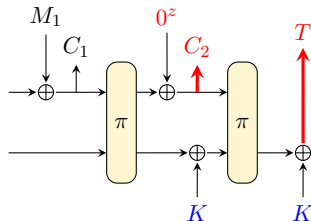
NEW ATTACKS

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Reasonable for suffix-ZP

- Attackers check **collision on $C_2 || T$ together**.
- Complexity: $2^{\frac{\tau+z}{2}}$.
- It matches the proof for suffix [NSS23, Theorem 1].



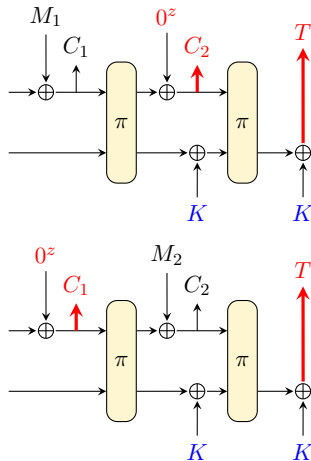
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Unreasonable for prefix-ZP

- 2S-TC₁ – **collision on C_1** .
- 2S-TC₂ – **collision on T** through brute-force.
- Complexity: $2^{\frac{z}{2}} + 2^{\tau+1}$.
- Counterexample of [NSS23, Theorem 2].





- 2S-TC outperforms TC if $2^{\frac{z}{2}} + 2^{\tau+1} < 2^{\frac{z+\tau}{2}}$, i.e., if $\max(\frac{z}{2}, \tau + 1) + 1 < \frac{z+\tau}{2}$.
- $\max(\frac{z}{2}, \tau + 1) = \frac{z}{2} \Rightarrow \frac{z}{2} + 1 < \frac{z+\tau}{2}$, for $2 < \tau$. So, if $4 < 2\tau + 2 < z$, 2S-TC wins.
- $\max(\frac{z}{2}, \tau + 1) = \tau + 1 \Rightarrow \tau + 2 < \frac{z+\tau}{2}$, for $\tau + 4 < z$. For $\tau + 4 < z \leq 2\tau + 2$, 2S-TC wins.
- So, in total 2S-TC **outperforms** the simple TC attack as long as $z > \tau + 4$.

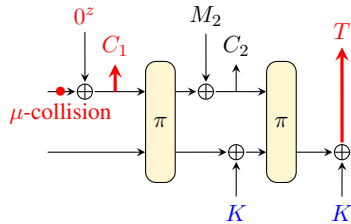
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Example of an instance

$$r_F = r_G = 128, c_F = c_G = 384, l = 256, \nu = 128, \kappa = 128, \tau = 128, z = 256.$$

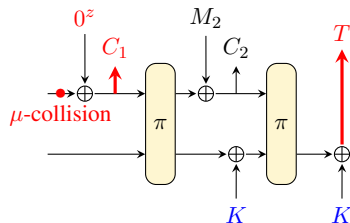
IS	CC	TC	2S-TC
$2^l = 2^{256}$	$2^{\min\{\frac{c_F}{2}, \frac{c_G+z}{2}\}} = 2^{192}$	$2^{\frac{\tau+z}{2}} = 2^{192}$	$2^{\frac{z}{2}} + 2^{\tau+1} = 2^{128} + 2^{129} \approx 2^{129.58}$

- We can generate μ -collision on C_1 .
- Collision on T becomes a collision search over μ -choices.
- Complexity: $2^{mulcoll(z,\mu)} + 2^{\tau+\log \mu}$.
- It can be faster than $2^{\frac{z}{2}} + 2^{\tau}$ by a factor upto $\log z$.





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IS	CC	TC	2S-TC	2S-TC ($\mu = 15$)
2^{256}	2^{192}	2^{128}	$2^{128} + 2^{64}$	$2^{125.26}$

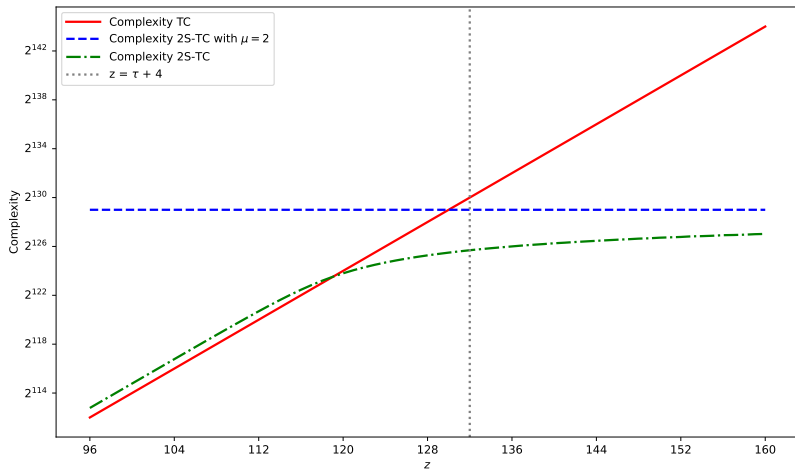
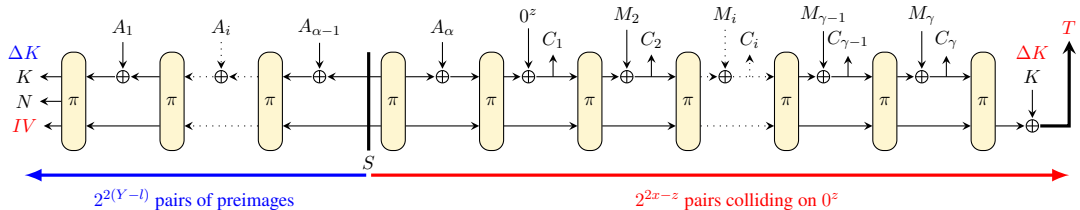
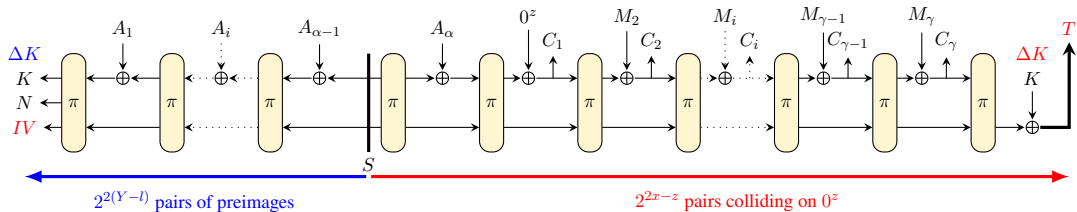


Figure: Complexity of different attacks for tag length $\tau = 128$ and varying zero-padding length $z \in [96, 160]$.

- Forward: find (A_α, M) and (A'_α, M') so that their ciphertext values collide and get ΔK .
- Backward: find $(A_{\alpha-1}, \dots, A_1)$ such that the IV matches and get same ΔK .
- Complexity: $2^{\frac{\tau+z}{4} + \frac{l}{2} + 1}$.



- Forward: find (A_α, M) and (A'_α, M') so that their ciphertext values collide and get ΔK .
- Backward: find $(A_{\alpha-1}, \dots, A_1)$ such that the IV matches and get same ΔK .
- Complexity: $2^{\frac{\tau+z}{4} + \frac{l}{2} + 1}$.



Example of an instance

$$r = r_F = r_G = 128, c = c_F = c_G = 384, l = 64, \nu = 192, \kappa = 256, \tau = 256, z = 128.$$

IS	CC	TC	MitM
$2^{l + \frac{\kappa}{2}} = 2^{192}$	$2^{\frac{c}{2}} = 2^{192}$	$2^{\frac{\tau+z}{2}} = 2^{192}$	$2^{\frac{\tau+z}{4} + \frac{l}{2} + 1} = 2^{96+32+1} = 2^{129}$



- We studied 3 attacks IS / CC / TC with KB / ZP w.r.t. CMT-4 / CMT-K^{*}.
- 2S-TC shows that Zero-padding is not just a longer tag.
- In the case of Zero-Padding, position of padding matters.
- Tag KB blocks some attacks, but can enable a new MitM attack.
- Committing security is parameter- and structure-sensitive.



<https://ia.cr/2026/1160>





Thank You!